

Tutorial 1: Oracle JDBC Development Environment

CSCI3170 Introduction to Database Systems

Learning Objectives

By the end of this tutorial, you will be able to:

- Explain the roles of the JDK, JDBC API, Oracle JDBC driver, and IntelliJ IDEA.
- Install a compatible Java Development Kit and IntelliJ IDEA.
- Create and run a Java project in IntelliJ IDEA.
- Add the Oracle JDBC driver (`ojdbc17.jar`) to a project.
- Verify that Java can load the Oracle JDBC driver.

Key Concept: The Four Pieces

- **JDK** compiles and runs Java programs.
- **JDBC** is the standard Java API for working with relational databases. It is included in the JDK, so there is no separate “JDBC application” to install.
- **Oracle JDBC driver** is a JAR file that implements JDBC for Oracle Database.
- **IntelliJ IDEA** is the development environment in which we write, run, and debug Java code.

All four must be configured correctly before a Java program can communicate with Oracle.

Part 1: Install IntelliJ IDEA and Java

For this course, use a **64-bit JDK 26**.

The easiest method is to let IntelliJ IDEA download the JDK when you create the project in Part 2. But the download speed can be slow. Alternatively, you can download a JDK 26 distribution from Oracle’s website. Oracle’s Java downloads are available at:

<https://www.oracle.com/java/technologies/downloads/>

After installation, open a terminal and verify both commands:

```
java -version
javac -version
```

Both commands should report version `26.x`. If `java` works but `javac` does not, you probably installed only a runtime rather than a JDK.

Step 2: Install IntelliJ IDEA

Download IntelliJ IDEA for your operating system from:

<https://www.jetbrains.com/idea/download/>

The free features are sufficient for this course. Accept the default installation options. On Windows, enabling the launcher and file-association options is convenient but not required.

Part 2: Create and Run a Java Project

Step 1: Create the Project

1. Start IntelliJ IDEA and choose **New Project**.
2. Select **Java** and name the project CSCI3170-Labs.
3. Select **IntelliJ** as the build system.
4. In the **JDK** field, select an installed JDK 26. If none is listed, choose **Download JDK**, select version **26** and a trusted vendor, and complete the download.
5. Disable **Add sample code**, then click **Create**.

Step 2: Run a Small Java Program

In the Project tool window, right-click the `src` folder and choose **New** → **Java Class**. Name it `HelloDatabase` and enter:

```
public class HelloDatabase {  
    public static void main(String[] args) {  
        System.out.println("CSCI3170 Java environment is ready.");  
    }  
}
```

Click the green Run icon beside `main`. The Run window should display:

CSCI3170 Java environment is ready.

Part 3: Install the Oracle JDBC Driver

Required Driver Version

Use the course-provided **Oracle 21c** `ojdbc17.jar` (or download the `ojdbc17.jar` listed under Oracle Database 21c). This driver supports JDK 26 and can communicate with the departmental Oracle 12.2 server. Do not automatically choose the newest 26ai driver, because its supported database versions may not include Oracle 12.2.

Step 1: Obtain the Driver

Use the copy provided on the course website. If no copy is provided, use Oracle's official JDBC download page:

<https://www.oracle.com/database/technologies/appdev/jdbc-downloads.html>

Download `ojdbc17.jar` and save it in a stable location. Do not unzip the JAR file.

Step 2: Add the Driver to IntelliJ IDEA

1. In IntelliJ IDEA, choose **File** → **Project Structure**.
2. Select **Libraries**, click **+**, and choose **Java**.
3. Select the downloaded `ojdbc17.jar`.
4. When asked, attach the library to the CSCI3170-Labs module.
5. Click **Apply**, then **OK**.

You should now see the driver under **External Libraries**. The JAR is a *project dependency*; merely downloading it is not enough.

Part 4: Verify the Driver

Create a Java class named `DriverCheck` and enter:

```
public class DriverCheck {
    public static void main(String[] args) {
        try {
            Class<?> driver = Class.forName("oracle.jdbc.OracleDriver");
            System.out.println("Oracle JDBC driver loaded: "
                               + driver.getName());
        } catch (ClassNotFoundException e) {
            System.err.println("Oracle JDBC driver was not found.");
            System.err.println("Add ojdbc17.jar to the module dependencies.");
        }
    }
}
```

Run the class. A correct setup prints:

```
Oracle JDBC driver loaded: oracle.jdbc.OracleDriver
```

Modern JDBC programs normally discover the driver automatically. We use `Class.forName` here only as a clear installation check.

Part 5: Troubleshooting Checklist

- **No JDK is listed:** Open **File** → **Project Structure** → **Project** and select or download JDK 26.
- **javac is not recognized:** Install a full JDK, then restart the terminal and IntelliJ IDEA.
- **ClassNotFoundException:** Confirm that `ojdbc11.jar` appears under both **Libraries** and the module's **Dependencies** tab.
- **UnsupportedClassVersionError:** Confirm the project SDK and Run configuration both use JDK 26.
- **Driver stored in Downloads:** Move it to a stable course folder, then update the library path. Moving or deleting the JAR later breaks the project.

Part 6: Checkpoint

1. Output of `java -version` and `javac -version`.
2. The CSCI3170-Labs project using JDK 26.
3. `ojdbc11.jar` under External Libraries.
4. Successful output from `DriverCheck`.

End of Tutorial 1. The lecturer will take over tutorial 2 with exercises on design theory. In Tutorial 3, we will connect Java to the departmental Oracle server.